

Foundations of Artificial Intelligence

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Exercise Sheet 1

Due: February 22, 2026

Please note the submission rules at the end of this exercise sheet. Non-adherence to these rules may lead to your submission not being marked.

Exercise 1.1 (1+1+1+1 marks)

Characterize the following AI systems with respect to the four categories (acting humanly, thinking humanly, thinking rationally, acting rationally) that have been introduced in the lecture. Justify your answers in one to two sentences.

- (a) Pepper ([https://en.wikipedia.org/wiki/Pepper_\(robot\)](https://en.wikipedia.org/wiki/Pepper_(robot)))
- (b) AlphaGo (<https://en.wikipedia.org/wiki/AlphaGo>)
- (c) ACT-R (<https://en.wikipedia.org/wiki/ACT-R>)
- (d) Google Maps (specifically for route planning; https://en.wikipedia.org/wiki/Google_Maps)

Note: In many cases several categories can be argued for, meaning there is not only one correct answer. Choose one category and present your argument for it.

Exercise 1.2 (1.5+1.5 marks)

- (a) Consider a self-driving taxi that takes the most scenic route instead of the fastest route. Does it behave rationally? Justify your answer.
- (b) Consider ChatGPT, a chatbot designed to be a digital assistant and answer questions. There are various examples where we find that ChatGPT provides incorrect answers and tricks users by trying to make the answer sound correct. Is this behavior rational?
Hint: ChatGPT is a language model designed to predict the most likely next words in a series of words and it was trained with a form of reinforcement learning that incorporates human feedback (see RLHF¹).

Exercise 1.3 (1.5+1.5 marks)

Consider an agent in a 2D grid where certain cells are blocked trying to reach the upper right corner of the grid. The agent can observe whether it is at the goal or not, and whether the cells north, south, east and west are blocked.

- (a) Provide a description of a possible implementation for such a reflexive agent.
- (b) Is the agent guaranteed to find the goal (if it is reachable) in finitely many steps? Justify your answer.

¹<https://openai.com/blog/deep-reinforcement-learning-from-human-preferences/>

Submission rules:

- Submit one solution in groups of up to two students, stating the names of all group members in the submission. All group members are equally responsible for all submitted exercises.
- Create a single A4 PDF file for all non-programming exercises. Solutions in L^AT_EX are preferred, but clearly legible scans in A4 with reasonable contrast and file size are accepted as well.
- In case there are programming exercises, upload a ZIP file (ending .zip, .tar.gz or .tgz; *not* .rar or anything else) including your PDF and those code textfiles required by the exercise. Put your names in a comment on top of each code file. Make sure your code compiles and test it. Code that does not compile or which we cannot successfully execute will not be graded.
- Include a statement on scientific integrity at the end of your submission (see below).

Statement on scientific integrity:

- All submitted solutions must be the product of your own work.
- If your written solutions were produced using any help (e.g., from peers or a large language model), list in what way these people or tools contributed to the solutions.
- We do not mark submissions that lack a statement on scientific integrity.
- Examples:
 - “We solved all exercises on our own.”
 - “We used ChatGPT to explain exercise 7 which we did not understand.”
 - “We had no time to solve this exercise sheet and copied all solutions from our friends.”
(If this is the case, thanks for your honesty but this is plagiarism, results in a warning and no marks. It is preferable to submit no solution at all.)